

ENERGY MANAGEMENT IN
MICROGRID
GRID CONNECTED MODE

DISTRIBUTION AUTOMATION & SMART GRID

EC9130

14TH WEEK PROGRESS TECHNICAL REPORT



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Contribution by the members in group

Tasks	2022/E/002	2022/E/093	2022/E/147
1.Three-phase grid system modelling		√	√
2.dq control and PLL implementation	√	√	
3. Simulation and result analysis	√		√
4. Progress Report making	√		√

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1. INTRODUCTION

Modern power systems are developing into smart grids, where renewable energy sources and the utility grid are connected together. In such systems, proper power conversion and synchronization are important to maintain stable operation. renewable sources like PV systems and batteries produce DC power, while the grid operates with AC power (400 V, 50 Hz). Therefore, a grid-connected inverter is required to convert DC power into AC power and supply it to the grid

The inverter must also match the grid in terms of frequency, phase angle, and voltage, and control the power flow properly. In this system, a Phase Locked Loop (PLL) is used to synchronize the inverter with the grid. A dq control method is used to control active and reactive power, where I_d controls active power and I_q controls reactive power. PI controllers are used to generate control signals. A PWM technique is used to generate switching pulses for the inverter, and a filter inductor is used to reduce harmonics and produce smooth current.

In Week 14, the three-phase grid-connected inverter system is designed and simulated. This work is important for connecting the PV and battery system to the grid in the next stage.

2. OBJECTIVES

2.1 Develop three-phase grid system

To model a three-phase grid system in Simulink with standard voltage and frequency. This is used as the reference system for connecting the inverter and analyzing its performance.

2.2 Implement PLL synchronization

To design a PLL that can detect the grid frequency and phase angle. This helps the inverter to synchronize properly with the grid and maintain stable operation.

2.3 Design dq control system

To implement dq transformation and control the inverter using PI controllers. In this method, I_d controls active power and I_q controls reactive power, making the system easier to control.

2.4 Implement PWM-based inverter control

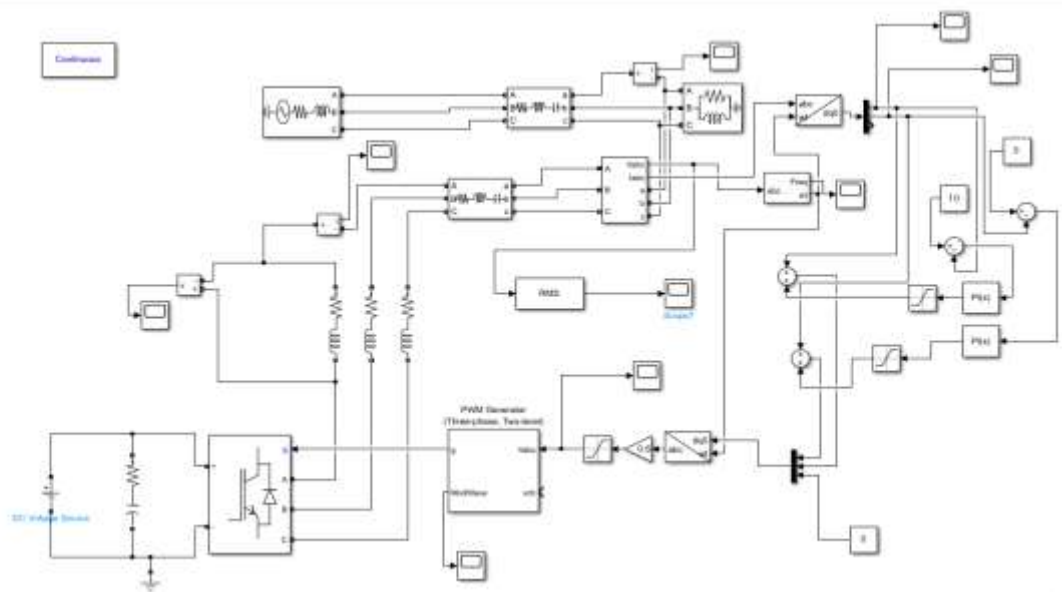
To generate switching signals for the inverter using PWM technique. This controls the output voltage and ensures proper conversion from DC to AC power.

3. SYSTEM CONFIGURATION

3.1 Overall Architecture

The system is designed as a three-phase grid-connected inverter system. It converts DC power into AC power and supplies it to the grid and load.

- DC source (600 V)
- Three-phase inverter
- Filter inductor
- Grid and load
- PLL and dq control system
- PWM generator



3.2 Three-Phase Grid System

✓ The grid is modeled as a balanced three-phase AC source with

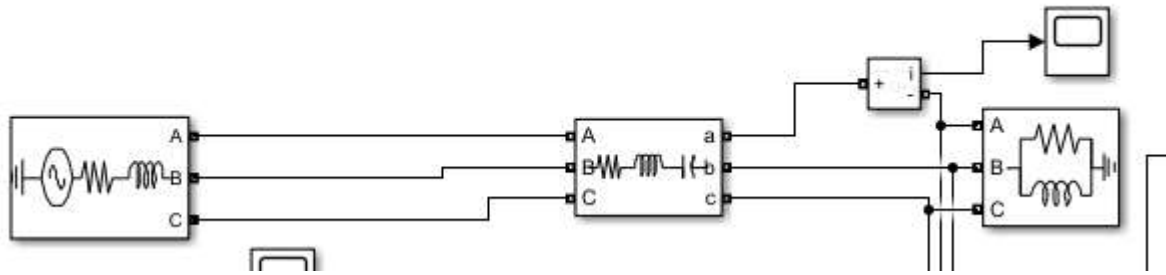
- Line-to-line voltage = 400 V
- Frequency = 50 Hz

The grid acts as a reference system, and the inverter must match its voltage and frequency. ensure constant power delivery.

▪ **Phase Voltage**

$$V_{\text{phase}} = \frac{400}{\sqrt{3}} \approx 230\text{V}$$

✓ This is the expected RMS voltage per phase.



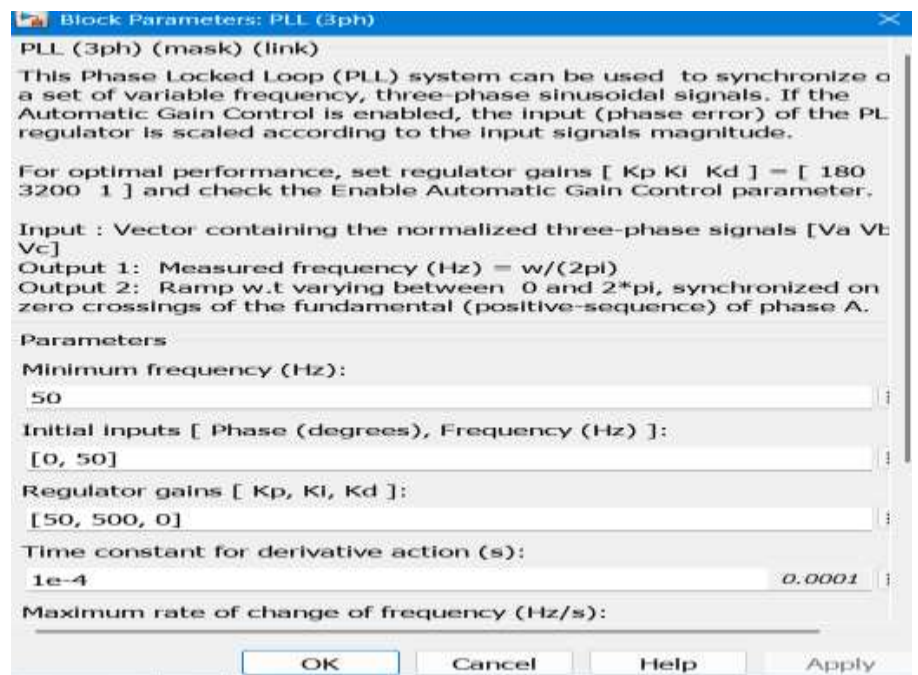
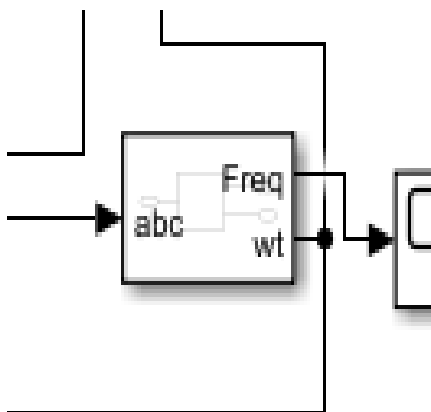
3.3 Phase Locked Loop

- ✓ The PLL is used to synchronize the inverter with the grid.

It measures the grid voltage and provides

- Phase angle (θ)
- Frequency = 50 Hz

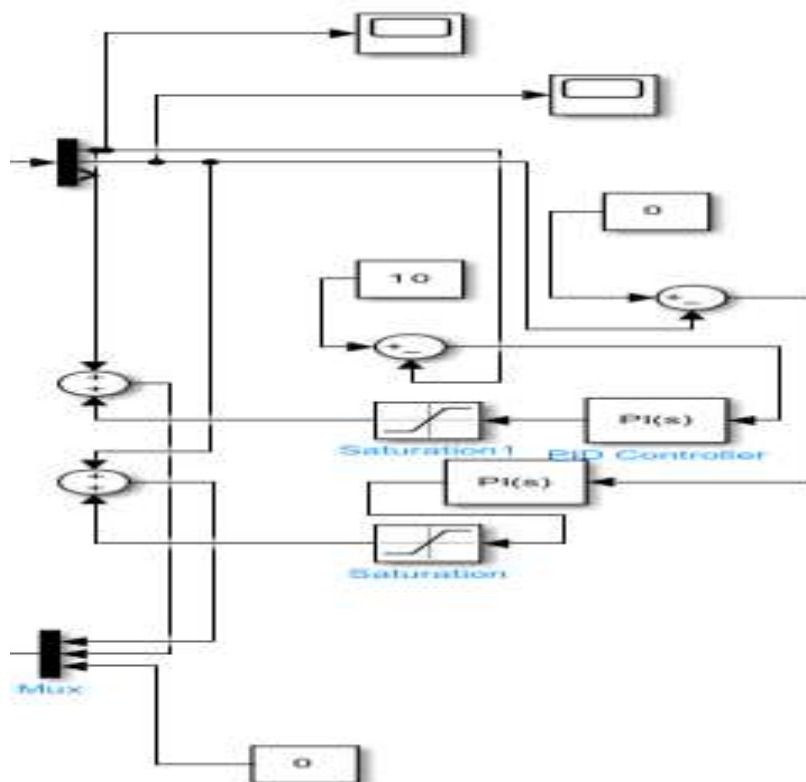
This information is used in dq transformation for proper control.



3.4 dq Control System

- ✓ The three-phase currents are converted into dq components using abc to dq transformation.
- I_d controls active power
- I_q controls reactive power

Reference values are compared with measured values, and PI controllers generate control signals to reduce error. Here In this system $\mathbf{Iq=0}$ because reactive power is very small.



3.5 PWM Generator and Inverter

- ✓ The PWM generator produces switching signals for the inverter.

The modulation index is limited to,

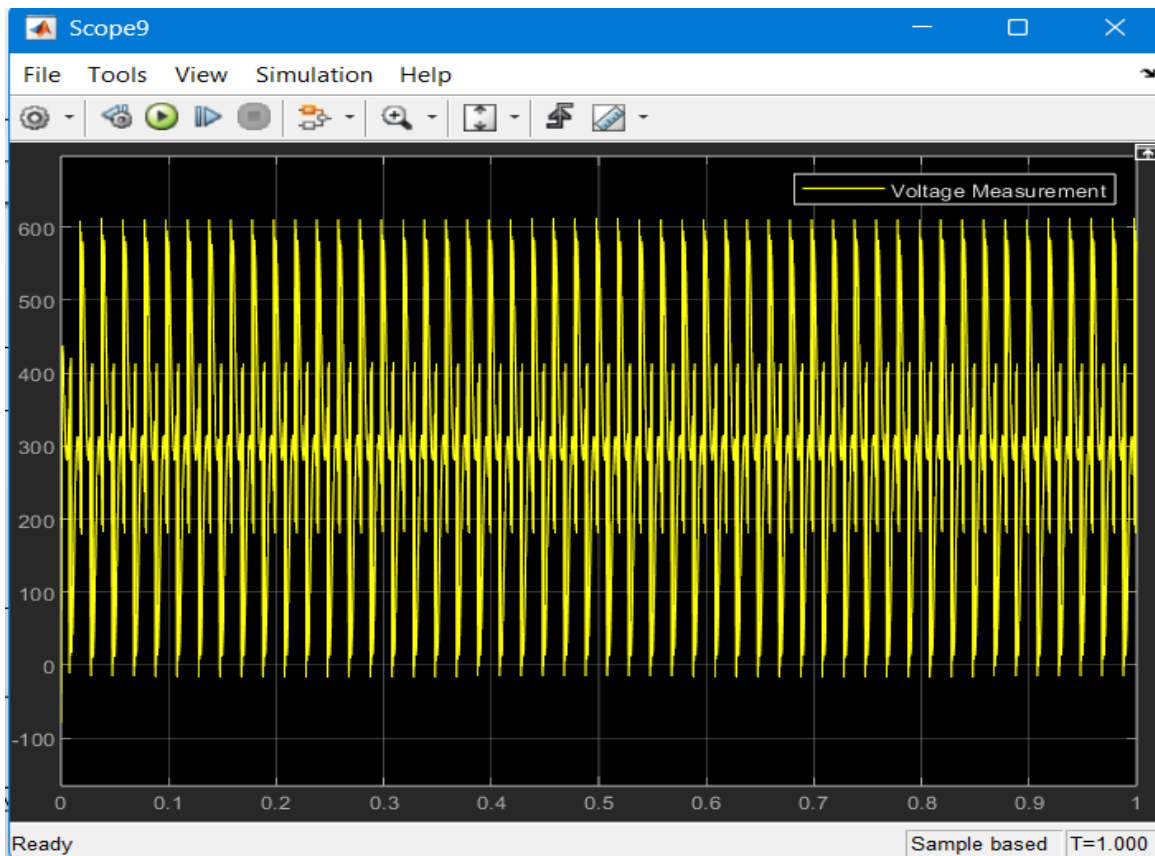
$$-1 \leq m \leq 1$$

This prevents over-voltage and ensures stable operation.

- ✓ The DC source provides a constant voltage of 600 V, which acts as the input to the inverter.

The inverter is a three-phase universal bridge, which converts DC voltage into AC voltage using switching device is IGBT. the inverter output is controlled by PWM signals to generate a sinusoidal AC waveform.

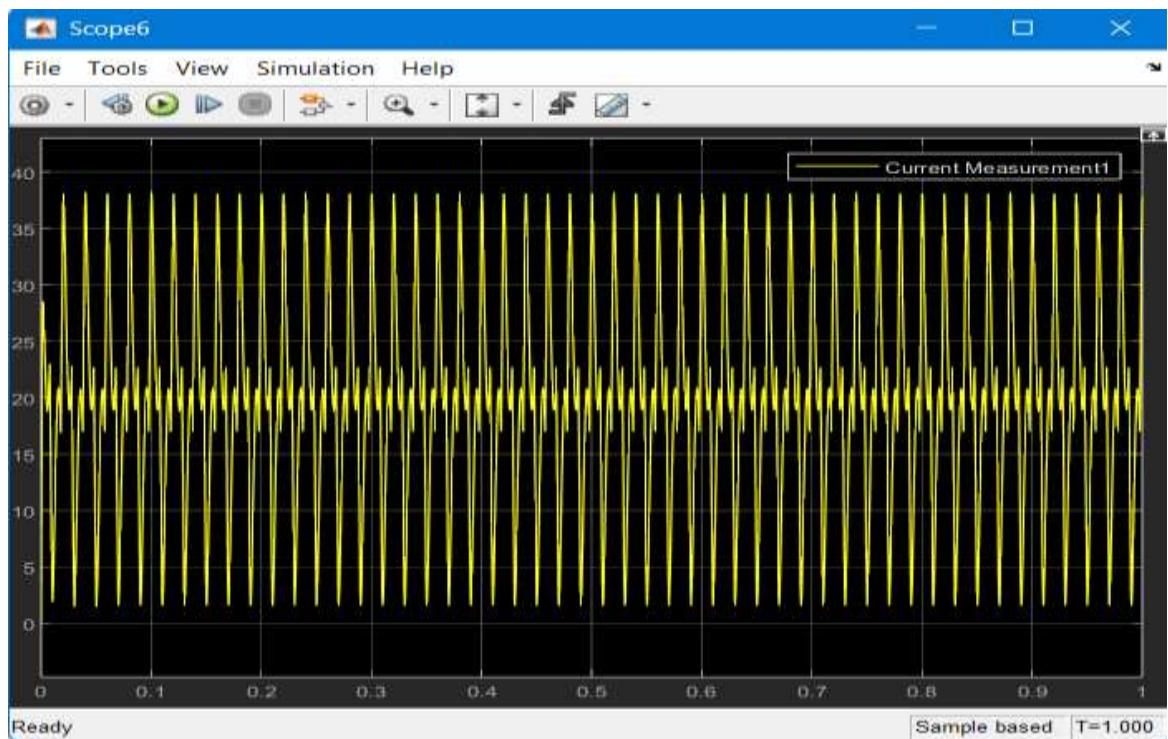
4. SIMULATION RESULTS AND ANALYSIS



The output voltage waveform shows a PWM switching waveform

Peak voltage ≈ 600 V

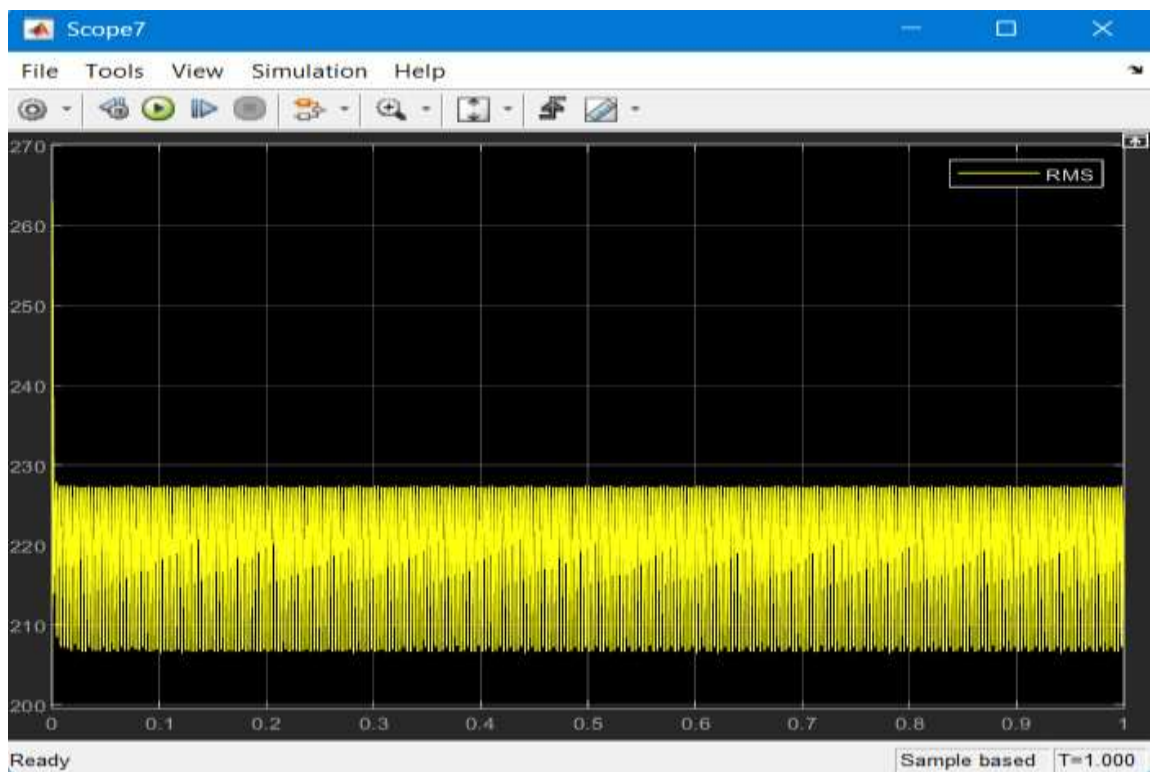
- Hence, Average AC voltage corresponds to grid level



The Current waveform is nearly sinusoidal

- Hence, Current magnitude = ± 20 A peak current, RMS current = 15 A

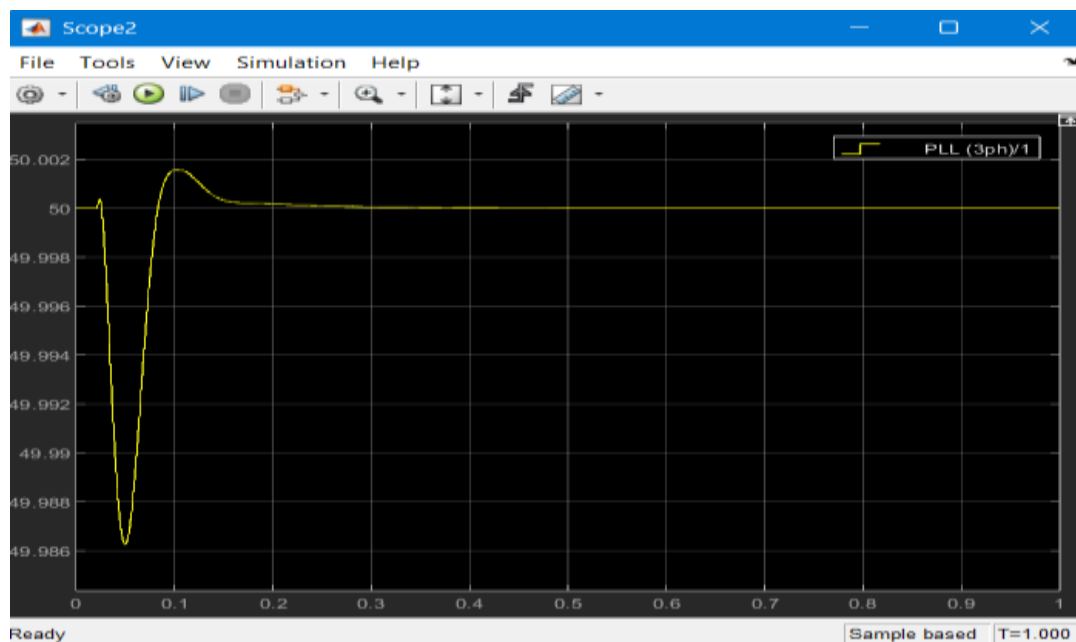
here, Small ripple is present due to switching, but overall waveform is stable.



✓ The RMS voltage = 220 – 230 V per phase

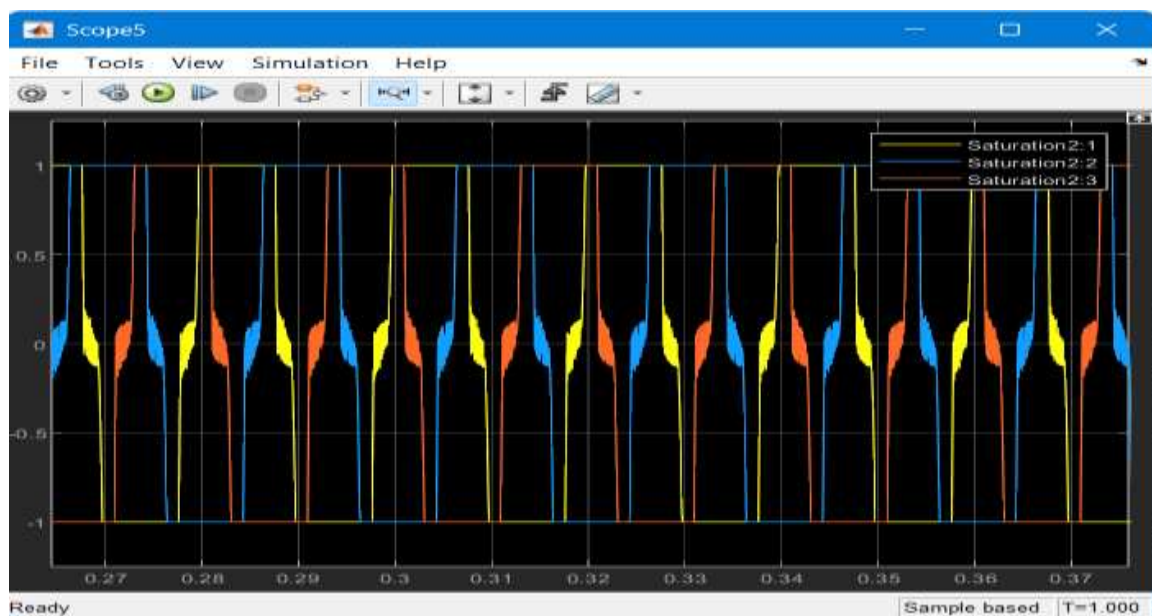
Theoretically expected value is,

$$V_{\text{phase}} = \frac{400}{\sqrt{3}} \approx 230\text{V}$$



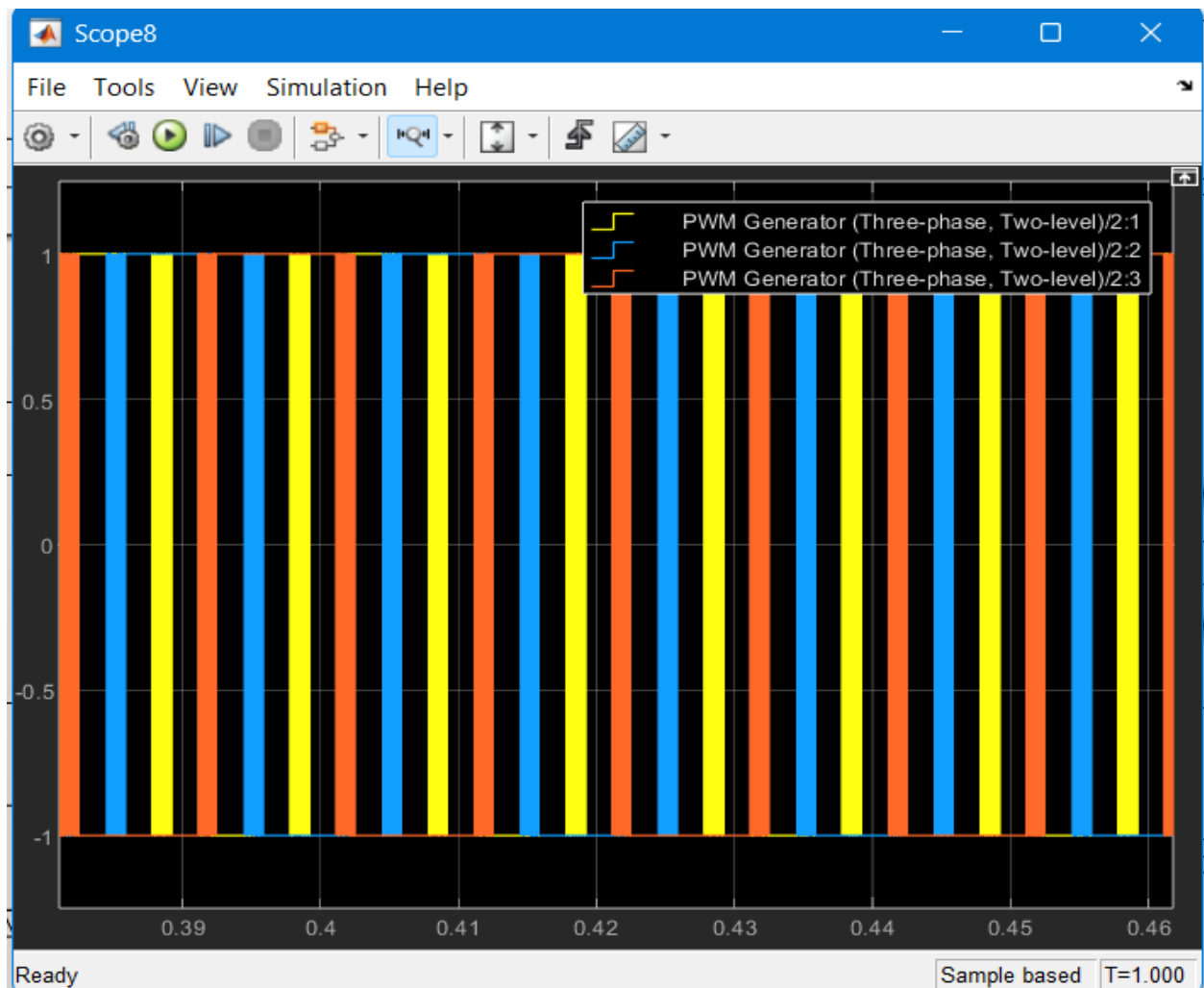
✓ The Frequency = 50 Hz

But, Small transient at starting, then stable
Here, indicates proper grid synchronization.



✓ Here, here, Signals are limited between -1 and 1

This is Confirms proper modulation control and no over-modulation



✓ Three-phase PWM pulses are observed

Accordingly signals are balanced and periodic so this is Ensures correct inverter switching operation.

5. DISCUSSION

In Week 14, we designed and simulated the three-phase grid connected inverter system. In this system, we used PLL, dq control, PWM generator and filter components. at the beginning, some problems occurred in the system. the output voltage was not stable because the modulation signal was high. due to that, the current also increased and system became unstable. to solve this problem, we added gain and saturation blocks to limit the modulation signal between -1 and 1. after that, the voltage and current became stable.

In the dq control section, at first the outputs were not coming properly in Scope 3 and Scope 4. This happened due to wrong signal connections and missing inputs from PLL. after correcting the connections and properly linking dq transformation with PLL output, the dq signals started working. we also analyzed the effect of filter resistance. when the resistance value is reduced, the current increases and waveform becomes more clear. when the resistance is increased, the current reduces and dq outputs become very small. this is because higher resistance limits the current flow in the circuit.

From the simulation results, the RMS voltage is around 230 V, which is the expected value. the current waveform is nearly sinusoidal with small ripple due to PWM switching. the PLL output shows stable frequency at 50 Hz, which means the system is synchronized with the grid. the dq current results show that I_d is almost constant and I_q is close to zero. this indicates that the system mainly supplies active power and reactive power is very small.

Finally, after fixing all errors and tuning the system, the inverter works properly. the voltage, current, and frequency are stable, and the system is ready for next stage integration.

6. CONCLUSION

In Week 14, the three-phase grid-connected inverter system was successfully designed and simulated. the system includes inverter modeling, PLL synchronization, dq control, and PWM generation. at the beginning, some errors occurred in the control section and signal connections. these issues were corrected by properly connecting the dq blocks and adding gain and saturation blocks to limit the modulation signal.

After these corrections, the system shows stable performance. the output voltage is maintained at the required level, and the current waveform is nearly sinusoidal with small ripple. the PLL ensures that the system operates at 50 Hz and stays synchronized with the grid. the dq control method works properly, where I_d controls active power and I_q is close to zero, indicating minimal reactive power.

Overall, the system operates correctly and is stable. this work is an important step for integrating the PV and battery system with the grid in the next stage.

7. REFERENCES

1. J. M. Guerrero et al., “Hierarchical control of droop-controlled AC and DC microgrids,” IEEE Transactions on Industrial Electronics, vol. 58, no. 1, pp. 158–172, Jan. 2011.
2. S. Pulenthirarasa, P. R. Satpathy, V. K. Ramachandaramurthy, A. Ramasamy, A. Atputharajah, and S. Padmanaban, “Adaptive energy management and control of an autonomous PV-wind-diesel-BESS hybrid microgrid: A case study on Tioman Island, Malaysia,” Results in Engineering, vol. 28, 2025, Art. no. 107731.